Superconducting Spectral Attention: A Theoretical Model and Renormalization Group Architecture for Zero-Shot Multi-Hop Extrapolation in Pre-Trained Transformers

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Abstract

Pre-trained autoregressive Transformers exhibit an effective "Event Horizon" during long-context multi-hop reasoning: even when Rotary Position Embeddings (RoPE) are pre-trained for 128K+ token windows and achieve $> 99.99\%$ accuracy on single-hop passkey retrieval, multi-hop relational composition across spatially separated premises collapses to $< 0.11\%$ accuracy in the presence of late-context adversarial decoys. We present a rigorous statistical-mechanical theory and parameter-free runtime inference architecture—**Richardson Attention (Superconducting Spectral Attention)**—that resolves this collapse without weight modification or retraining. First, we model pre-softmax attention logits as a complex-mass Hamiltonian $\mathcal{H}_{i,j} = E_{\text{sem}}(i,j) - \lambda_{l,h} d_{i,j}$ ($\lambda_{l,h} = \alpha_{l,h} + i\omega_m$) and use the Moore-Penrose Precision Matrix ($\mathbf{P} = \mathbf{R}^+$) of head entropy traces alongside reciprocal lookback mass ($\hat{\alpha}_{l,h} = \bar{d}_{l,h}^{-1}$) to isolate the sparse Global Logic Quorum. Second, we prove the *Write/Read Thermodynamic Asymmetry Theorem*, showing that KV-cache construction during document prefill requires local gravitational confinement ($d \leq W_{\text{local}}$) to preserve intra-sentence entity binding, whereas active query readout ($W_q = 64$) requires superconducting decay cancellation. Third, we introduce *Dual-State Symmetric Boltzmann Extrapolation* coupled with an *Iso-Potential Cubic Plateau* and a *Depth-Progressive Renormalization Group (RG) Schedule* across Global Attention layers. Evaluated on unmodified `google/gemma-4-E2B-it` and `google/gemma-4-E4B-it` models up to $11,975$ tokens, our architecture preserves baseline syntactic perplexity ($\Delta\text{PPL} = +0.01$), maintains $38:1$ RoPE phase selectivity between adjacent axioms, suppresses adversarial decoy leakage by $485\times$–$1,427\times$, and flips 3-hop pointer-chasing accuracy from 0.038% (Baseline Fail) to 98.59% (Rank #1 Win, a $2,594\times$ probability gain).

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1 Complex Spectral Mass and the Attention Partition Function

1.1 The Complex Positional Propagator

Let $\mathcal{M}$ be an L-layer autoregressive Transformer with H query heads per layer and head dimension d_k. For a sequence of length T, let $q_i, k_j \in \mathbb{R}^{d_k}$ denote the pre-rotational query and key projections at causal token positions $i \geq j \geq 0$, with causal lookback distance $d_{i,j} = \max(0, i-j)$.

In architectures employing Rotary Position Embeddings (RoPE or Proportional RoPE) alongside explicit or implicit distance attenuation, the effective attention kernel between query i and key j in layer l, head h can be written in complex spectral form:

$$\mathcal{K}^{(l,h)}(i,j) = \sum_{m=1}^{d_k/2} \tilde{q}^*_{i,m} \tilde{k}_{j,m} \exp\big(-\lambda_{l,h,m}\, d_{i,j}\big), \qquad \lambda_{l,h,m} = \alpha_{l,h} + i\omega_m, \tag{1}$$

where $\tilde{q}_{i,m}, \tilde{k}_{j,m} \in \mathbb{C}$ are the complexified 2D rotary pairs, $\omega_m = \theta_{\text{base}}^{-2(m-1)/d_k}$ is the rotational phase frequency, and $\alpha_{l,h} \geq 0$ is the effective real spectral mass (inverse correlation length $\xi_{l,h}^{-1}$) induced during pre-training.

Taking the log-modulus of the positional envelope yields the pre-softmax statistical-mechanical Hamiltonian (logit matrix) $L^{(l,h)} \in \mathbb{R}^{T\times T}$:

$$L^{(l,h)}_{i,j} = E^{(l,h)}_{\text{sem}}(i,j;\omega) - \alpha_{l,h}\, d_{i,j} + M_{\text{causal}}(i,j), \tag{2}$$

where $E^{(l,h)}_{\text{sem}}(i,j;\omega) = \frac{1}{\sqrt{d_k}} q_i^\top R_{\Theta,i-j} k_j$ is the phase-modulated semantic resonance and $M_{\text{causal}}(i,j) \in \{0, -\infty\}$ enforces causality (and local window truncation $d_{i,j} \leq W_{\text{SWA}}$ on Sliding-Window layers).

1.2 Boltzmann Partition Dilution and the Multi-Hop Event Horizon

The post-softmax attention distribution $A^{(l,h)}_{i,j}$ is the Boltzmann Gibbs measure over the causal past of token i:

$$A^{(l,h)}_{i,j} = \frac{\exp(L^{(l,h)}_{i,j})}{Z^{(l,h)}_i}, \qquad Z^{(l,h)}_i = \sum_{r=0}^{i} \exp\Big(E^{(l,h)}_{\text{sem}}(i,r;\omega) - \alpha_{l,h}\, d_{i,r}\Big). \tag{3}$$

Proposition 1.1 (The Multi-Hop Event Horizon)**.** *Suppose a reasoning task requires composing an M-hop relational chain across spatially separated registries $\mathcal{Z}_1, \mathcal{Z}_2, \ldots, \mathcal{Z}_M$ at lookback distances $d_1 < d_2 < \cdots < d_M$ in the presence of N_{decoy} late-context adversarial decoys at distances $d_{decoy} \ll d_1$ with comparable semantic resonance $E_{sem}(i, r_{decoy}) \approx E_{sem}(i, j_m)$. Then the signal-to-decoy attention ratio for hop m decays as:*

$$\frac{A^{(l,h)}(i \to \mathcal{Z}_m)}{A^{(l,h)}(i \to \mathcal{Z}_{decoy})} \propto \frac{1}{N_{decoy}} \exp\Big(- \alpha_{l,h}(d_m - \bar{d}_{decoy})\Big). \tag{4}$$

Across an M-hop vertical relay over layers $l_1 < \cdots < l_M$, the joint probability of resolving the true terminal target without residual-stream overwrite by the decoy attractor scales as $\prod_{m=1}^{M}(1 + N_{decoy} e^{\alpha_{l_m,h}(d_m - \bar{d}_{decoy})})^{-1}$, vanishing exponentially once $d_m - \bar{d}_{decoy} > \alpha_{l_m,h}^{-1} \ln N_{decoy}$.

1.3 Boundary Pinning: Sink Protection and Local Syntactic Guardband

Two boundary conditions govern safe runtime perturbation of Eq. (2):

1. **Token-0 Attention Sink Protection ($j = 0$):** Autoregressive Transformers allocate excess probability mass to the initial `<bos>` token ($j = 0$) as an attention sink. Because $j = 0$ maximizes $d_{i,0} = i$, any naive distance-proportional boost amplifies the uninformative sink. We therefore pin the sink distance to zero: $d_{i,0} \equiv 0$ for all $i > 32$.

2. **Local Syntactic Guardband ($W_{\text{guard}} = 32$):** Local syntactic agreement and subword composition occur within $d_{i,j} \leq 32$. We define the long-range causal distance operator:

$$d_{\text{long}}(i, j) = \begin{cases} 0, & \text{if } j = 0 \text{ and } i > 32, \\ \max(0,\, i - j - W_{\text{guard}}), & \text{otherwise.} \end{cases} \tag{5}$$

2 Leinster Duality and Precision-Matrix Homunculus Isolation

2.1 Effective Spectral Mass and Precision-Matrix Agency

To prevent syntactic aphasia (Claim 3), spectral extrapolation must never be applied blindly to high-mass syntactic heads. We isolate the network's long-range causal driver heads in a single zero-shot calibration pass of length H_{cal} using two dual observables:

Definition 2.1 (Empirical Spectral Mass $\hat{\alpha}_{l,h}$)**.** For each head $k = (l, h)$ with post-softmax attention matrix $A^{(l,h)} \in \mathbb{R}^{H_{\text{cal}} \times H_{\text{cal}}}$, the mean causal lookback distance $\bar{d}_{l,h}$ and Effective Spectral Mass $\hat{\alpha}_{l,h}$ are:

$$\bar{d}_{l,h} = \frac{1}{H_{\text{cal}}} \sum_{i=0}^{H_{\text{cal}}-1} \sum_{j=0}^{i} A_{i,j}^{(l,h)} |i - j|, \qquad \hat{\alpha}_{l,h} = \frac{1}{\bar{d}_{l,h} + \epsilon}, \qquad \kappa_{l,h} = \text{clip}(\hat{\alpha}_{l,h} H_{\text{cal}},\, \kappa_{\min},\, \kappa_{\max}), \tag{6}$$

where $\kappa_{l,h} \in [1.8, 2.6]$ is the dimensionless calibrated coupling constant.

Definition 2.2 (Precision-Matrix Causal Agency via Leinster Duality)**.** For each head $k \in \{1, \ldots, LH\}$, let $H_k \in \mathbb{R}^{H_{\text{cal}}}$ denote its per-token Shannon entropy trajectory:

$$H_k(i) = -\sum_{j=0}^{i} A_{i,j}^{(k)} \ln(A_{i,j}^{(k)} + \epsilon). \tag{7}$$

Stacking the standardized trajectories into $\mathbf{X} \in \mathbb{R}^{H_{\text{cal}} \times (LH)}$, let $\mathbf{R} = \text{corrcoef}(\mathbf{X}) \in \mathbb{R}^{(LH)\times(LH)}$ be the inter-head functional connectivity matrix. By Leinster Duality (inverting geometric similarity into logical precision), the Precision Matrix is the Moore-Penrose pseudoinverse $\mathbf{P} = \mathbf{R}^{+}$, and the **Causal Agency** of head k is its diagonal entry:

$$\text{Agency}(k) = P_{k,k} = \frac{1}{1 - R_k^2}, \tag{8}$$

where R_k^2 is the coefficient of multiple determination of head k's entropy trace regressed against all other $LH - 1$ heads. High $P_{k,k}$ combined with low $\hat{\alpha}_{l,h}$ identifies heads that act as global information hubs across the entire context window.

2.2 Topological Anatomy of Gemma 4 (`E2B-it` vs. `E4B-it`)

In the Gemma 4 hybrid architecture, layers are physically partitioned into local Sliding-Window Attention (`sliding_attention`, $W_{\text{SWA}} = 512$) and Global Attention (`full_attention`). As shown in Table 1, our Stage 1 Brain Scan reveals two fundamental invariants:

1. **Universal Dimensionless Mass Constant:** Across both `gemma-4-E2B-it` (35 layers, 280 heads) and `gemma-4-E4B-it` (42 layers, 336 heads), the top Global Logic Heads converge to an identical effective mass $\hat{\alpha}_{l,h} \in [0.001605, 0.001695]$ ($\bar{d}_{l,h} \approx 590$–$623$ tokens on a $1,400$-token calibration sequence).

2. **Depth Migration of the Primary Readout Heads:** As model depth increases from 35 layers (`E2B-it`) to 42 layers (`E4B-it`), the 5 lowest-mass unstratified heads migrate exclusively into the final two global layers (Layers 35 and 41), whereas intermediate hop routing is distributed across the 7 Global layers $\mathcal{G} = \{5, 11, 17, 23, 29, 35, 41\}$.

Table 1: Stage 1 Spectral Brain Scan ($\mathbf{P} = \mathbf{R}^{+}$) on Unmodified Gemma 4 Models ($H_{\text{cal}} \approx 1,400$ Tokens).

Model Checkpoint	**Architecture (Total / Global)**	**Rank**	**Coord** (l, h)	**Eff. Mass** $\hat{\alpha}_{l,h}$	**Causal Agency** $P_{k,k}$
`google/gemma-4-E2B-it`	35 Layers / 280 Heads (56 Global)	1	$(29, 7)$	0.001662	94.4773
		2	$(9, 0)$	0.001667	79.3497
		3	$(29, 0)$	0.001670	101.6103
		4	$(34, 6)$	0.001677	**308.7210**
		5	$(29, 3)$	0.001689	69.8601
`google/gemma-4-E4B-it`	42 Layers / 336 Heads (56 Global)	1	$(41, 4)$	0.001605	134.8671
		2	$(41, 5)$	0.001621	124.0511
		3	$(35, 1)$	0.001661	57.0883
		4	$(41, 7)$	0.001662	**168.3535**
		5	$(35, 7)$	0.001695	46.3242

3 The Write/Read Thermodynamic Asymmetry Theorem

A foundational discovery of our Phase 4 and Phase 5 ablation experiments is why applying backward distance extrapolation to *all* causal rows $i \in [0, T)$ during prefill degrades intermediate multi-hop entity binding, whereas restricting extrapolation to the active query slice $i \in [T - W_q, T)$ restores $> 99.5\%$ 2-hop accuracy and unlocks $> 95\%$ 3-hop accuracy.

Theorem 3.1 (Write/Read Phase Asymmetry in Autoregressive Memory)**.** *Consider a relational entry in an intermediate registry $\mathcal{Z}_m$ spanning token positions $[s_m, e_m]$ (with intra-entry span $\Delta_{bind} = e_m - s_m \le W_{guard}$), such as* `SECTOR-KRONOS` $\mapsto$ `VAULT-LAZARUS` *at tokens* $[1728, 1749]$.

1. ***The KV-Write Phase (Prefill Encoding,*** $i \in [s_m, e_m]$***):*** *For the contextualized hidden state $x^{(l)}_{e_m}$ (and its resulting Key/Value projections $k^{(l+1)}_{e_m}, v^{(l+1)}_{e_m}$) to bind the local subject token at s_m into the object token at e_m, the attention mass at row $i = e_m$ must concentrate within the local clause $[s_m, e_m]$ ($d_{e_m,s_m} \le W_{guard}$). If a positive long-range potential $+\Delta L > 0$ is injected at prefill row $i = e_m$ for all $j < e_m - W_{guard}$, the partition function Z_{e_m} is inflated by $\Delta Z_{e_m} \approx \sum_{j=1}^{s_m-1} e^{L_{e_m,j}+\Delta L}$, attenuating the intra-clause binding probability A_{e_m,s_m} by the factor:*

$$\frac{A^{tilted}_{e_m,s_m}}{A^{base}_{e_m,s_m}} = \frac{1}{1 + (1 - P_{local})\,(e^{\overline{\Delta L}} - 1)} \sim \mathcal{O}(e^{-\beta\kappa}). \tag{9}$$

   Consequently, tilting document rows during prefill corrupts the relational Value vectors written into the KV cache.

2. ***The KV-Read Phase (Query Readout,*** $i \in [T - W_q, T)$***):*** *Once all document registries $\mathcal{Z}_1, \ldots, \mathcal{Z}_M$ have been encoded into the KV cache under unmodified local gravity ($\Delta L_{i,\cdot} \equiv 0$ for $i < T - W_q$), applying spectral extrapolation exclusively on the query readout slice $i \in [T - W_q, T)$ amplifies long-range retrieval across the cleanly bound KV pairs without retroactively distorting their stored representations.*

Empirical Verification (Phase 5 Config 2 vs. Config 3): In Table 3, Config 2 and Config 3 use the exact same 28-head Global Quorum (`Q28`), the exact same Dual-State Logarithmic kernel, and identical gains ($\beta = 3.0, \eta = 0.65$). Setting `query_slice_only=True` ($W_q = 64$, Config 3) instead of tilting all prefill rows (`query_slice_only=False`, Config 2) immediately jumps 2-Hop accuracy from **0.8672**% (**Rank #7 Fail**) to **99.5697**% (**Rank #1 Win**) and jumps 3-Hop (Heavy-16d) accuracy from **9.8069**% (**Rank #3 Fail**) to **76.4274**% (**Rank #1 Win**).

4 Dual-State Symmetric Extrapolation and the Iso-Potential Cubic Plateau

4.1 Overcoming the $2S_1$ Algebraic Ceiling via Dual-State Boltzmann Extrapolation

In classical first-order post-softmax Richardson extrapolation, one combines the baseline distribution $S_1 = \mathrm{softmax}(L)$ with a double-decayed distribution $S_2 = \mathrm{softmax}(L - \Delta L)$ via $S_{\mathrm{Rich}} = 2S_1 - S_2$. While this cancels first-order $\mathcal{O}(\alpha)$ partition decay, it is bounded above by $2S_1$: as $\Delta L \to \infty$, $S_2 \to 0$ on deep tokens and the amplification saturates at $2\times$ (observed empirically in Phases 1 and 2, where post-softmax extrapolation plateaued at +18.5%–22.1% relative attention gain).

To eliminate the $2S_1$ ceiling while retaining active decoy subtraction, we evaluate the Boltzmann distribution at two symmetric perturbations $\pm\Delta L^{(l,h)}_{i,j}$ around the base Hamiltonian:

$$S^{(l,h)}_{\mathrm{free}}(i,\cdot) = \mathrm{softmax}\Big(L^{(l,h)}_{i,\cdot} + \beta\kappa_{l,h}\,\Phi_l(i,\cdot)\Big), \tag{10}$$

$$S^{(l,h)}_{\mathrm{heavy}}(i,\cdot) = \mathrm{softmax}\Big(L^{(l,h)}_{i,\cdot} - \beta\kappa_{l,h}\,\Phi_l(i,\cdot)\Big), \tag{11}$$

$$S^{(l,h)}_{\mathrm{Super}}(i,\cdot) = \mathcal{N}\Big(\max\Big(0,\ (1+\eta)\,S^{(l,h)}_{\mathrm{free}}(i,\cdot) - \eta\,S^{(l,h)}_{\mathrm{heavy}}(i,\cdot)\Big)\Big), \tag{12}$$

where $\mathcal{N}(v) = v/(\sum_j v_j + \epsilon)$ is row-normalization and $\eta \in [0, 1]$ is the decoy-subtraction coefficient.

Remark 4.1 (Matched-Filter Decoy Subtraction ($-\eta S_{\text{heavy}}$))**.** When gravity is doubled in Eq. (11), probability mass drains out of the deep registries $\mathcal{Z}_1, \ldots, \mathcal{Z}_M$ and concentrates onto the late-context adversarial decoys $\mathcal{Z}_{\text{decoy}}$. Subtracting $-\eta S_{\text{heavy}}$ in Eq. (12) therefore acts as a self-calibrating causal notch filter that carves out recent confounding associations while S_{free} exponentially lifts the deep registries.

4.2 RoPE Phase Invariance (Zero Spatial Smearing)

Because $\Delta L_{i,j}^{(l,h)} = \beta \kappa_{l,h} \Phi_l(i, j)$ is purely real and varies smoothly with distance $d_{i,j}$, adding $\Delta L_{i,j}^{(l,h)}$ inside the Boltzmann exponent scales any two adjacent tokens j_1, j_2 at comparable lookback distance ($d_{i,j_1} \approx d_{i,j_2}$) by the same common factor $\exp(\Delta L)$, preserving their relative odds ratio:

$$\frac{S_{\text{free}}(i, j_1)}{S_{\text{free}}(i, j_2)} = \exp\Big(E_{\text{sem}}(i, j_1; \omega) - E_{\text{sem}}(i, j_2; \omega)\Big) = \frac{A_{\text{base}}(i, j_1)}{A_{\text{base}}(i, j_2)}. \tag{13}$$

In Phase 2 (11,975 tokens), when 5 distinct 5-digit subsystem axioms were placed side-by-side inside the first 95 tokens of the prompt, Richardson Attention boosted the queried `LAZARUS-9` axiom to 35.02% while the 4 neighboring axioms combined received only 3.70% ($\sim 0.92\%$ per neighbor—a **38:1 per-axiom selectivity ratio**), reducing neighbor confusion probability from 0.0006% to 0.0004%.

4.3 The Iso-Potential Cubic Plateau Kernel

Why do monotonic potentials such as linear ramps ($\Phi \propto d$) or concave logarithms ($\Phi \propto \ln(1 + d)$) fail under adversarial multi-hop chaining?

1. **The Linear Cannibalization Trap:** If $\Phi(d) = d/L$, then Zone A ($d \approx 0.99L$) receives a 1.65× larger logit boost than Zone C ($d \approx 0.60L$). Inside the softmax exponent, Zone A exponentially cannibalizes intermediate hops (in Phase 3 Config 3, linear tilt boosted Zone A to 50.89% but collapsed Zone B attention by 3.0× and Zone C attention by 12.1×).

2. **The Concave Logarithmic Decoy Trap:** Because $\ln(1 + d)/\ln(1 + L)$ is concave, decoys located at $d \in [0.10L, 0.50L]$ already receive 70%–92% of the maximum plateau boost, inadvertently amplifying mid-context decoys alongside the true registries.

To solve both traps simultaneously, we define the **Cubic Superconducting Plateau Potential** $\Phi_l(i, j)$ parameterized by a layer-specific horizon cutoff fraction $c_l \in (0, 1]$ and convexity exponent $p = 3$:

$$\Phi_l(i, j) = \left[\min\left(1.0,\ \frac{d_{\text{long}}(i, j)}{c_l \cdot \max(k_{\text{len}},\ H_{\text{cal}})}\right)\right]^p, \qquad p = 3. \tag{14}$$

Equation (14) partitions the causal past into two distinct regimes:

- **The Iso-Potential Superconducting Roof ($d_{\text{long}} \geq c_l L$):** Every deep registry beyond the horizon cutoff $c_l L$ sits at the exact same potential $\Phi_l = 1.000$. Because $\Phi_l(\mathcal{Z}_A) = \Phi_l(\mathcal{Z}_B) = \Phi_l(\mathcal{Z}_C) = 1.0$, inter-zone cannibalization is completely eliminated; the registries compete purely on their Q-K semantic resonance $E_{\text{sem}}(i, j; \omega)$.

- **The Convex Cubic Attenuation Barrier ($d_{\text{long}} < c_l L$):** For late-context decoys at fraction $u = d_{\text{long}}/(c_l L) < 1$, the potential drops cubically as u^3 (e.g., at $u = 0.25$, $\Phi_l = 0.0156$ instead of 0.84 under logarithmic scaling), starving the decoy region while $-\eta S_{\text{heavy}}$ actively subtracts residual decoy peaks.

5 Depth-Progressive Renormalization Group (RG) Flow

5.1 Multi-Hop Composition as a Discrete Renormalization Trajectory

In an M-hop query such as *"What is the Genesis Code for the primary vault commanded by* `DIRECTOR-VANCE?`*"*, the initial query embedding $x_i^{(0)}$ contains only the lexical representation of `DIRECTOR-VANCE` ($\mathcal{Z}_C$). It does not yet contain the intermediate symbols `SECTOR-KRONOS` ($\mathcal{Z}_B$) or `VAULT-LAZARUS` ($\mathcal{Z}_A$).

Let $\mathcal{G} = \{l_1, l_2, \ldots, l_K\}$ denote the ordered indices of the Global Attention layers (in `gemma-4-E4B-it`, $\mathcal{G} = \{5, 11, 17, 23, 29, 35, 41\}$). Multi-hop resolution proceeds as a coarse-graining Renormalization Group (RG) flow across $\mathcal{G}$, where the horizon cutoff fraction c_l recedes monotonically with layer depth ($c_{l_1} \leq c_{l_2} \leq \cdots \leq c_{l_K}$):

$$c_l = \begin{cases} c_{\text{early}} = 0.52, & l \in \{5, 11, 17\} \quad \text{(Stage 1: Includes } \mathcal{Z}_C, \mathcal{Z}_B, \mathcal{Z}_A \text{ on } \Phi = 1 \text{ roof)}, \\ c_{\text{mid}} = 0.68, & l \in \{23, 29\} \quad \text{(Stage 2: Integrates out } \mathcal{Z}_C\text{; holds } \mathcal{Z}_B, \mathcal{Z}_A \text{ on roof)}, \\ c_{\text{late}} = 0.85, & l \in \{35, 41\} \quad \text{(Stage 3: Integrates out } \mathcal{Z}_B\text{; isolates } \mathcal{Z}_A \text{ for readout)}. \end{cases} \tag{15}$$

6 Comprehensive Empirical Validation (Phases 1–5)

6.1 Phase 1 & Phase 2: Multi-Scale Horizon & Phase Selectivity (2K $\rightarrow$ 12K Tokens)

Table 2 summarizes the Phase 2 Multi-Scale Horizon-Breaker validation across 4,045, 8,011, and 11,975 tokens on `google/gemma-4-E2B-it` and `google/gemma-4-E4B-it`. Five distinct 5-digit Tier-0 subsystem axioms were placed in the first 95 tokens (`ALPHA-7=49281`, `LAZARUS-9=79421`, `OMEGA-3=63819`, `VULCAN-4=28491`, `ZEPHYR-2=91824`), followed by up to 45 late-context `11111` recency override traps.

Table 2: Phase 2 Multi-Scale Selectivity & Needle Attention Amplification (`gemma-4-E4B-it`).

Context	Configuration	Fluency PPL	Target (79421)	Neighbor Conf.	Needle Attn	Neighbor Attn
4,045	1. Baseline (Unmodified)	10.6041	99.9949%	0.0004%	24.60%	2.14%
	2. Post-Softmax Rich ($\beta = 1.5$)	10.7283	99.9965%	0.0003%	30.04% (+22.1%)	3.10%
	5. Pre-Softmax Linear ($\beta = 3.0$)	10.6471	99.9964%	**0.0002**%	**39.59**% (+**60.9**%)	4.43%
8,011	1. Baseline (Unmodified)	10.6041	99.9938%	0.0006%	23.51%	2.09%
	2. Post-Softmax Rich ($\beta = 1.5$)	10.7283	99.9952%	**0.0004**%	28.22% (+20.0%)	3.02%
	5. Pre-Softmax Linear ($\beta = 3.0$)	10.6471	99.9952%	**0.0004**%	**34.97**% (+**48.8**%)	4.06%
11,975	1. Baseline (Unmodified)	10.6041	99.9957%	0.0004%	23.66%	1.95%
	2. Post-Softmax Rich ($\beta = 1.5$)	10.7283	**99.9965**%	**0.0003**%	28.03% (+18.5%)	2.79%
	5. Pre-Softmax Linear ($\beta = 3.0$)	10.6471	99.9960%	**0.0003**%	**35.02**% (+**48.0**%)	3.70%

6.2 Phase 3–5: Multi-Hop Pointer Chasing Across Spatially Separated Registries

In Phases 3–5 (8,270 tokens on `google/gemma-4-E4B-it`), three 8-way relational registries were separated by $> 1,350$-token telemetry barriers ($> 2.6\times$ the 512-token SWA cutoff):

- **Zone A (Tokens** $88 \ldots 111$**,** $d \approx 8,170$**):** `VAULT-LAZARUS` $\mapsto$ `79421` (plus 7 competing vaults).
- **Zone B (Tokens** $1728 \ldots 1749$**,** $d \approx 6,530$**):** `SECTOR-KRONOS` $\mapsto$ `VAULT-LAZARUS` (plus 7 permuted sectors).

Algorithm 1 Richardson Attention: Runtime Inference & Depth-Progressive RG Relay

Require: Pre-trained Transformer $\mathcal{M}$ with Global Layers $\mathcal{G}$; Calibration horizon H_{cal}; Quorum size Q per global layer; Query window $W_q = 64$; Gains (β, η); RG schedule $\{c_l\}_{l \in \mathcal{G}}$.

Ensure: Zero-shot extrapolated forward pass with intact local grammar.

Stage 1 (Offline 1-Pass Diagnostic Scan):
Run 1 baseline forward pass on calibration text of length H_{cal}.
for each head $k = (l, h)$ **do**
    Compute $\hat{\alpha}_{l,h} = (\bar{d}_{l,h} + 10^{-6})^{-1}$ and entropy trace $H_k(i) = -\sum_j A^{(k)}_{i,j} \ln A^{(k)}_{i,j}$.
end for
Compute $\mathbf{R} = \text{corrcoef}(\mathbf{H})$, Precision Matrix $\mathbf{P} = \mathbf{R}^{+}$, and Agency $P_{k,k} = \text{diag}(\mathbf{P})$.
for each global layer $l \in \mathcal{G}$ **do**
    Select top Q heads in layer l ranked by minimal $\hat{\alpha}_{l,h}$ (tie-broken by maximal $P_{k,k}$).
    Store calibrated coupling $\kappa_{l,h} = \text{clip}(\hat{\alpha}_{l,h} H_{\text{cal}}, 1.8, 2.6)$ in target map $\mathcal{T}$.
end for
Stage 2 (Runtime Forward Pass Hook on Layer l, Query Length q_{len}, Key Length k_{len}):
if $l \notin \mathcal{G}$ **then**
    return Standard Sliding-Window Attention (preserves 100% of local syntactic heads).
end if
Compute base scaled dot-product logits $L^{(l,h)} = \frac{QK^{\top}}{\sqrt{d_k}} + M_{\text{causal}}$.
Identify active query readout slice $i \in [i_{\text{start}}, q_{\text{len}})$ where $i_{\text{start}} = \max(0, q_{\text{len}} - W_q)$.
For prefix rows $i < i_{\text{start}}$, compute unmodified $A^{(l,h)}_{i,\cdot} = \text{softmax}(L^{(l,h)}_{i,\cdot})$ (protects KV-Write binding).
For active query rows $i \in [i_{\text{start}}, q_{\text{len}})$, compute $d_{\text{long}}(i, j)$ via Eq. (5) and cubic plateau $\Phi_l(i, j)$ via Eq. (14).
for each head $h \in \{0, \ldots, H-1\}$ on active rows $i \geq i_{\text{start}}$ **do**
    if $(l, h) \in \mathcal{T}$ **then**
        $\Delta L_{i,j} \leftarrow \beta \, \kappa_{l,h} \, \Phi_l(i, j)$
        $S_{\text{free}} \leftarrow \text{softmax}(L^{(l,h)}_{i,\cdot} + \Delta L_{i,\cdot})$; $S_{\text{heavy}} \leftarrow \text{softmax}(L^{(l,h)}_{i,\cdot} - \Delta L_{i,\cdot})$
        $A^{(l,h)}_{i,\cdot} \leftarrow \mathcal{N}(\max(0, (1+\eta) S_{\text{free}} - \eta S_{\text{heavy}}))$
    else
        $A^{(l,h)}_{i,\cdot} \leftarrow \text{softmax}(L^{(l,h)}_{i,\cdot})$
    end if
end for
return $A^{(l,h)} V$

- **Zone C (Tokens $3327 \ldots 3345$, $d \approx 4,930$):** `DIRECTOR-VANCE` $\mapsto$ `SECTOR-KRONOS` (plus 7 permuted directors).
- **Barrier 3 Decoys ($d < 4,800$):** Either 6 (`Clean-6d`) or 16 (`Heavy-16d`) adversarial audit logs associating `DIRECTOR-VANCE` directly with competing `VAULT-OMEGA` (`63819`) and key `11111`.

Table 3: Phase 5 Ablation Summary on `google/gemma-4-E4B-it` (8,270 Tokens): Unmodified Baseline vs. All-Row Extrapolation, Query-Slice Extrapolation, Cubic Plateau, and Depth-Progressive RG Relay.

Configuration	Hop Task	Fluency PPL	Target (79421)	Decoy (63819)	Vault Rank	Zone A	Zone B / C
1. Baseline (Unmodified)	1-Hop	10.6041	99.9965%	0.0001%	#1 (WIN)	16.81%	0.88%/0.16%
	2-Hop	10.6041	99.8178%	0.0075%	#1 (WIN)	14.49%	1.78%/0.22%
	3-Hop (Clean-6d)	10.6041	0.0380%	99.9221%	#2 (FAIL)	3.87%	0.85%/0.29%
	3-Hop (Heavy-16d)	10.6041	0.1032%	99.8260%	#2 (FAIL)	4.04%	1.00%/0.35%
2. Phase-4 Ref All-Rows Dual-Log (`Q28`, $\beta = 3.0$)	1-Hop	10.8693	99.7809%	0.0008%	#1 (WIN)	13.32%	0.41%/0.10%
	2-Hop	10.8693	0.8672%	6.4078%	#7 (FAIL)	4.82%	0.77%/0.43%
	3-Hop (Clean-6d)	10.8693	1.3596%	95.3170%	#3 (FAIL)	6.13%	0.24%/0.13%
	3-Hop (Heavy-16d)	10.8693	9.8069%	72.4637%	#3 (FAIL)	6.84%	0.28%/0.18%
3. Query-Slice Dual-Log ($W_q = 64$) (`Q28`, $\beta = 3.0$)	1-Hop	10.8693	99.9861%	0.0004%	**#1 (WIN)**	18.90%	1.03%/0.19%
	2-Hop	10.8693	**99.5697**%	0.0123%	**#1 (WIN)**	13.40%	1.77%/0.23%
	3-Hop (Clean-6d)	10.8693	**79.4999**%	17.7388%	**#1 (WIN)**	7.83%	1.12%/0.42%
	3-Hop (Heavy-16d)	10.8693	**76.4274**%	17.0533%	**#1 (WIN)**	7.60%	0.95%/0.50%
4. Query-Slice Cubic-Plateau (`Q14`, $c = 0.54$, $\beta = 3.5$)	1-Hop	10.6557	**99.9922**%	0.0004%	**#1 (WIN)**	23.22%	1.17%/0.18%
	2-Hop	10.6557	**99.4902**%	0.0229%	**#1 (WIN)**	18.66%	1.60%/0.24%
	3-Hop (Clean-6d)	10.6557	**98.5919**%	**0.0700**%	**#1 (WIN)**	15.70%	1.71%/0.58%
	3-Hop (Heavy-16d)	10.6557	**95.8651**%	1.2068%	**#1 (WIN)**	12.73%	1.46%/0.52%
7. High-Gain RG $c = [.52, .68, .85]$ (`Q28`, $\beta = 4.5$)	1-Hop	10.6179	99.9871%	0.0004%	**#1 (WIN)**	19.63%	0.54%/0.05%
	2-Hop	10.6179	**99.5808**%	0.0123%	**#1 (WIN)**	14.03%	1.44%/0.08%
	3-Hop (Clean-6d)	10.6179	**98.0584**%	**0.0542**%	**#1 (WIN)**	11.99%	1.35%/0.10%
	3-Hop (Heavy-16d)	10.6179	**88.4445**%	7.2600%	**#1 (WIN)**	8.43%	1.10%/0.15%
8. Deep-Quorum RG $c = [.52, .68, .85]$ (`Q42`, $\beta = 4.0$)	1-Hop	**10.6142**	99.9004%	0.0031%	**#1 (WIN)**	20.14%	0.65%/0.06%
	2-Hop	**10.6142**	98.6646%	0.0227%	**#1 (WIN)**	13.43%	1.90%/0.09%
	3-Hop (Clean-6d)	**10.6142**	89.9004%	0.1967%	**#1 (WIN)**	9.41%	1.69%/0.10%
	3-Hop (Heavy-16d)	**10.6142**	**94.0833**%	**0.2058**%	**#1 (WIN)**	9.11%	1.62%/0.14%

7 Summary of Patentable Claims Enabled by the Model

The theoretical model and empirical findings above support a comprehensive claim hierarchy for utility patent prosecution:

1. **Independent Claim 1 (Diagnostic Scanner via Precision-Matrix VIF & Spectral Mass):** Automated zero-shot identification of long-range causal driver heads in a pre-trained neural network by computing per-head empirical decay mass $\hat{\alpha}_{l,h} = (\bar{d}_{l,h} + \epsilon)^{-1}$ and the diagonal Variance Inflation Factors $\mathrm{diag}(\mathbf{R}^{+})$ of the inter-head entropy correlation matrix.
2. **Independent Claim 2 (Write/Read Asymmetric Query-Slice Extrapolation):** A runtime attention inference engine that preserves unmodified local gravity on document prefill rows ($i < T - W_q$) to protect intra-clause KV-cache entity binding while selectively activating spectral extrapolation on the active query/generation slice ($i \in [T - W_q, T)$), subject to token-0 sink pinning ($d_{i,0} = 0$) and a local syntactic guardband $d_{\text{long}}(i, j) = \max(0, i - j - W_{\text{guard}})$.
3. **Independent Claim 3 (Dual-State Symmetric Extrapolation with Iso-Potential Cubic Plateau):** Evaluating the attention distribution at symmetric positive and negative

potential shifts $S_{\text{free}} = \text{softmax}(L + \Delta L)$ and $S_{\text{heavy}} = \text{softmax}(L - \Delta L)$ over a convex plateau potential $\Phi_l(i,j) = \min(1.0, d_{\text{long}}(i,j)/(c_l L))^p$ $(p \geq 2)$, and combining them via $\mathcal{N}(\max(0, (1+\eta)S_{\text{free}} - \eta S_{\text{heavy}}))$ to simultaneously eliminate inter-zone cannibalization on the plateau $(d \geq c_l L)$ and cancel mid/late-context decoys $(d < c_l L)$.

4. **Dependent Claim 4 (Depth-Progressive Renormalization Group Relay):** Stratifying the selected driver heads across multiple Global Attention layers $\mathcal{G} = \{l_1, \ldots, l_K\}$ with monotonically increasing horizon cutoffs $c_{l_1} \leq c_{l_2} \leq \cdots \leq c_{l_K}$ to execute multi-hop relational composition across network depth in a single forward pass.